

SPARC: Single-Pass Scaling for Motion Forecasting with Conformal Bayesian Last Layers

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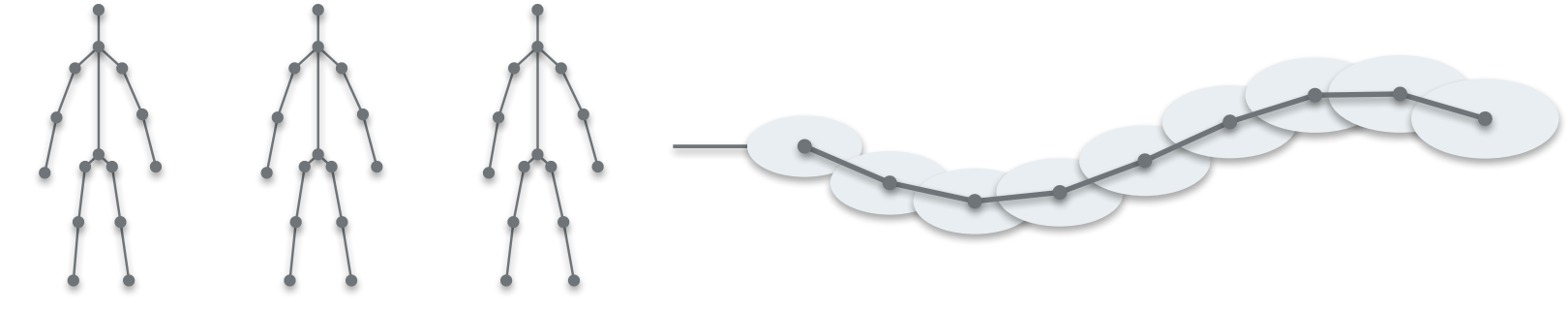
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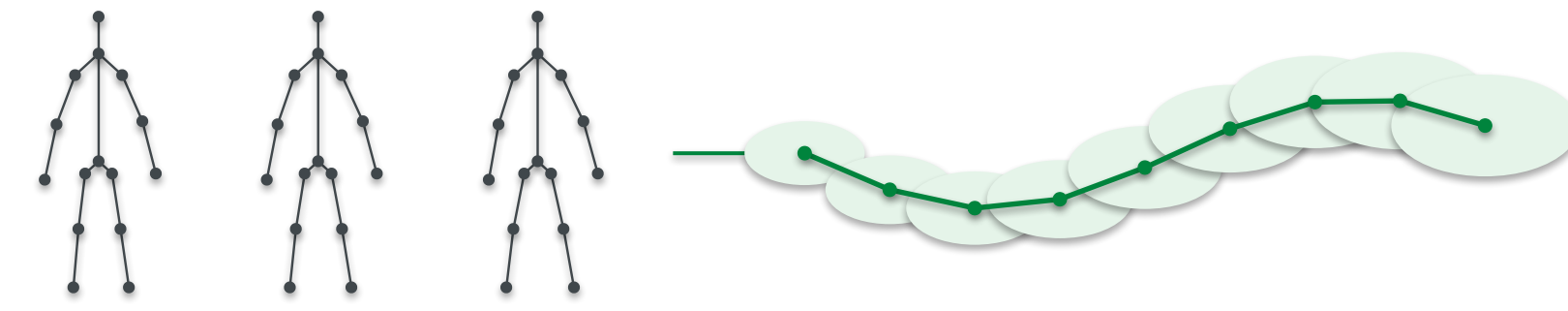
Why uncertainty?

Human motion forecasting predicts future 3D joint trajectories from a short observed prefix. A point forecast hides intrinsic ambiguity and failures outside familiar training support [2,11].

Point forecast / overconfident tube



Calibrated 95% marginal tube



Research question. Can a fast point forecaster become a structured, calibrated density without stochastic test-time sampling?

Why existing tools fall short

- **Repeated inference.** MC dropout and deep ensembles require stochastic passes [3,4].
- **Factorized outputs.** Analytic last-layer methods are efficient, but commonly use diagonal or isotropic residual models [5].
- **Calibration alone.** Conformal prediction gives marginal validity, but not an epistemic signal or trajectory covariance [8,9].

What SPARC contributes

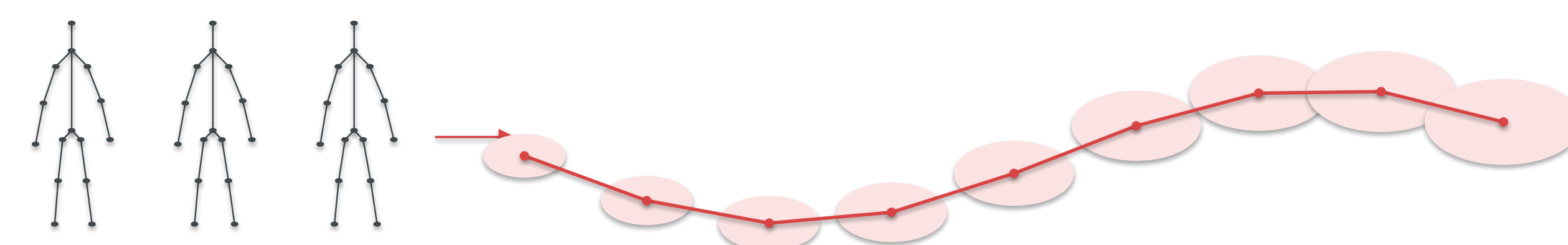
- 1 **Exact factorization.** $\kappa_t(x)\Sigma_{\text{str},t}(x)$ preserves the learned correlation structure.
- 2 **Unified recipe.** Bayesian leverage, graph-temporal covariance, and split CP in one layer.
- 3 **Broad evidence.** Nine dataset/protocol blocks with density, point, and tube metrics.
- 4 **Risk monitoring.** High- κ windows concentrate larger errors; thresholds remain held-out policies.

Qualitative intervals (3DPW)

Low- κ example: familiar feature region



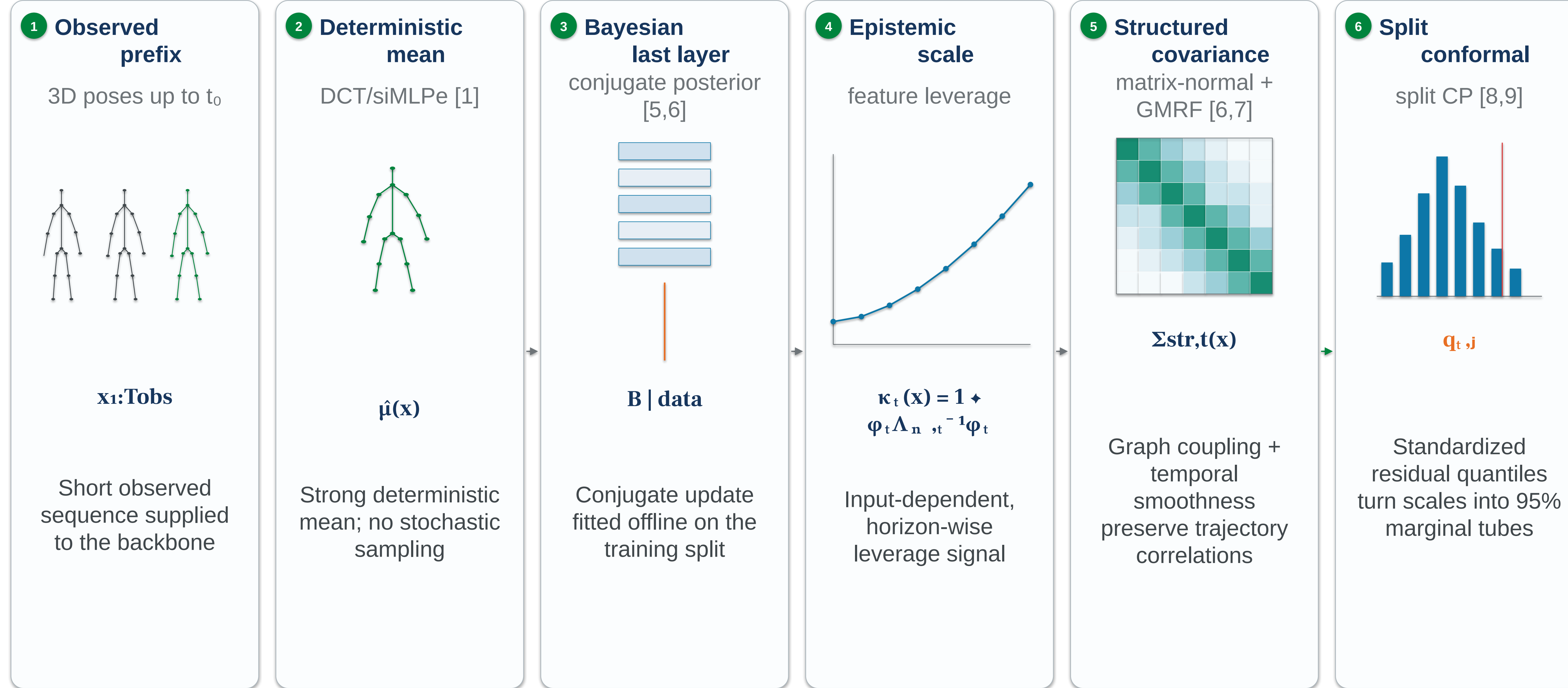
High- κ example: weak feature support



Low/high κ marks familiar/weak feature support. The structured density expands before both cases are conformalized to the same nominal marginal coverage.

SPARC: structured scale + conformal calibration in one pass

Starting from the DCT/siMLPe backbone [1], SPARC keeps the deterministic mean unchanged and adds analytic Bayesian leverage, graph-temporal covariance, and post-hoc split conformal calibration.



Proposition 1: exact structured-covariance scaling

$$\kappa_t(x) \cdot \Sigma_{\text{str},t}(x)$$

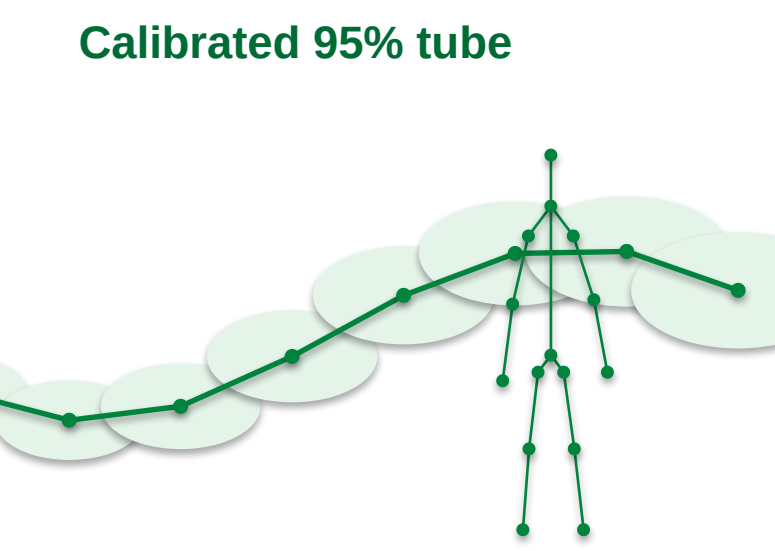
Under matrix-normal conjugacy, integrating the last-layer weights changes covariance magnitude, not correlations; the result holds for any positive-definite $\Sigma_{\text{str},t}$ [6].

Offline fit + calibration

1. Train: fit the mean, Bayesian last layer, and graph-temporal covariance.
2. Calibrate: store standardized residual quantiles on a disjoint split.
3. Freeze all parameters before the final evaluation split.

Online: one forward pass

1. Predict the mean and backbone features.
 2. Compute κ_t and scale the structured covariance.
 3. Apply the stored quantile to obtain a 95% tube.
- No sampling or test-time optimization.



Results across 9 dataset/protocol blocks

1.00	2.69	2.50	1.79×
NLL mean rank	MPJPE+NLL mean rank	W95(CP) mean rank	MPJPE in top-k decile

SPARC is not the uniform point-error winner. It instead leads density quality and the combined point+density trade-off while retaining efficient, calibrated uncertainty tubes.

Evaluation protocol. H36M plus seven cross-domain datasets; typically $T=50$ and $H=25$. Train, tuning, calibration ($n=512$), and evaluation ($n=1024$) are disjoint; $\alpha=.05$. Mean ranks normalize heterogeneous units.

NLL mean rank ($\downarrow$ better)

SPARC	1.00
Bridge-CQR [12]	3.00
Deep ensemble [4]	5.56
MeanFT [1]	6.11
CoMusion [13]	6.44
SPARD [14]	6.56

MPJPE + NLL mean rank ($\downarrow$ better)

SPARC	2.69
Bridge-CQR [12]	3.81
MeanFT [1]	4.17
CoMusion [13]	6.25
SPARD [14]	6.83
Deep ensemble [4]	6.89

Ready for deployment

1 One forward pass	95% Calibrated tubes	κ Risk monitor
No MC or diffusion sampling loop.	Finite-sample marginal validity [8,9].	Top-decile filtering lowers MPJPE by $\approx 9\%$.

Guarantee

$$\Pr\{Y_{t,j} \in \hat{C}_{t,j}(X)\} \geq 1 - \alpha$$

For a fresh exchangeable sample and each joint j / horizon t , split conformal gives finite-sample marginal validity [8,9].

Calibration is post hoc and disjoint: no model or covariance parameter is updated.

Scope and limitations

- Marginal per joint/horizon, not simultaneous over a full trajectory.
- Strong conditional guarantees require extra assumptions [10].
- κ is representation-dependent: learned features determine leverage.
- Arbitrary shift can violate exchangeability [11].

Use κ as a held-out policy signal, not as a theorem-backed alarm.

Paper & contact



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